

Experiment No: 1

Title:

Implement the program of substitution and transposition ciphers

Aim:

To implement substitution and transposition ciphers for hiding sensitive data and to understand their role in providing basic confidentiality in an Online Examination System.

Capstone Scenario – Online Examination System

In an online examination system, exam questions, exam codes, and student identifiers must not be visible in plain text during storage or transmission. Classical ciphers can be used to obfuscate data and demonstrate the importance of cryptography. In this experiment Substitution cipher or Transposition cipher is used to secure the question paper text. This simulates data hiding during early-stage secure exam processing.

Theory:

- X.800 Security Architecture
- Model for Network Security
- Symmetric Encryption Model
- Substitution Techniques and its types with Example
- Transposition Techniques and its type with Example

Algorithm

Algorithm 1 – Substitution Cipher (Caesar Cipher)

1. Read the plaintext (exam code).
2. Read the key value.
3. For each character:
 - Convert character to numeric value.
 - Add key value.
 - Apply modulo operation.
4. Convert back to character.
5. Display encrypted and decrypted text.

Algorithm 2 – Transposition Cipher

1. Read the plaintext (exam question).
2. Decide number of columns.
3. Write plaintext row-wise into a matrix.
4. Read matrix column-wise to generate ciphertext.
5. Reverse process for decryption.

Result

Thus, substitution and transposition ciphers were successfully implemented and tested for data hiding in an online examination system. The experiment demonstrated how basic cryptographic techniques provide confidentiality and laid the foundation for understanding advanced encryption algorithms used in secure online applications.

Conclusion

This experiment helps students:

- Understand classical cryptography
- Learn key-based encryption
- Appreciate the need for stronger encryption mechanisms

Experiment No: 2

Title:

Implement a Program to Perform Encryption and Decryption using DES / AES Cipher

Aim:

To implement DES and AES encryption and decryption algorithms for securing sensitive data and to understand their role in providing strong confidentiality in an Online Examination System.

Capstone Scenario – Online Examination System

In an online examination system, question papers, student responses, evaluation data, and result files are highly sensitive and must be protected against unauthorized access. Unlike classical ciphers, modern cryptographic algorithms are required to ensure practical security.

In this experiment:

- AES/DES encryption is used to encrypt exam question papers before storage or transmission.
- The encrypted data is decrypted only by authorized users using the correct secret key.

This simulates secure exam paper preparation, storage, and distribution in a real-world online examination system.

Theory:

- DES Architecture
- DES Single Round Structure
- Key Generation
- S-Boxes
- Advantages & Limitations
- AES Architecture
- AES Single Round Structure
- Key Generation Details of Every Steps
- Advantages & Limitations

Algorithm

1. Algorithm – AES Encryption and Decryption
2. Read the plaintext file (exam question paper).
3. Generate or read the secret key.
4. Initialize AES cipher in encryption mode.
5. Encrypt the plaintext to generate ciphertext.
6. Store or display encrypted data.
7. Initialize AES cipher in decryption mode.
8. Decrypt ciphertext using the same secret key.
9. Display the original plaintext.

Result

Thus, the DES and AES encryption and decryption algorithms were successfully implemented and tested for securing exam question papers in an online examination system. The experiment demonstrated how modern symmetric encryption algorithms provide strong confidentiality compared to classical ciphers.

Conclusion

This experiment helps students to:

- Understand modern symmetric key encryption
- Learn practical implementation of AES/DES using Java
- Realize the limitations of classical ciphers
- Appreciate the role of encryption in real-world secure systems

Experiment No: 3

Title:

Implement a program using Diffie Hellman key exchange algorithm

Aim:

To implement the Diffie–Hellman key exchange algorithm for securely generating a shared secret key and to understand its role in secure key distribution in an Online Examination System.

Capstone Scenario – Online Examination System

In an online examination system, encryption algorithms such as AES require a secret key. However, sharing this secret key over a network may lead to interception by attackers.

In this experiment:

- Diffie–Hellman key exchange is used to securely generate a shared secret key between the exam server and the exam client.
- This key is later used for encrypting exam papers and student responses.

This simulates secure session key establishment before encrypted exam data transmission.

Theory:

- Introduction
- Algorithm
- Example
- Man-In-The-Middle Attack

Algorithm

Algorithm – Diffie–Hellman Key Exchange

1. Agree on public values p and g .
2. Client generates private key a .
3. Server generates private key b .
4. Client computes public value $A = g^a \bmod p$.
5. Server computes public value $B = g^b \bmod p$.

6. Exchange A and B.
7. Client computes shared key $K = B^a \bmod p$.
8. Server computes shared key $K = A^b \bmod p$.
9. Verify both keys are identical

Result

Thus, the Diffie–Hellman key exchange algorithm was successfully implemented and tested for secure key generation in an online examination system, ensuring safe encryption key establishment.

Conclusion

This experiment helps students to:

- Understand secure key exchange
- Learn session key generation
- Prepare for SSL and encrypted communication

Experiment No: 4

Title:

Implement the program on Hash functions –SHA, MD5 to show the integrity check

Aim:

To implement cryptographic hash functions and to verify data integrity of exam-related files in an Online Examination System.

Capstone Scenario – Online Examination System

In an online examination system:

- Exam papers must not be altered
- Student answers must remain unchanged
- Result data must be verified

In this experiment:

- Hash values are generated for exam question papers
- Hash comparison is used to detect tampering

This simulates integrity verification during exam data transfer and storage.

Theory:

- HASH Function and Its Requirement with Symmetric and Asymmetric
- Encryption MAC Function and Its Requirement with Symmetric and Asymmetric Encryption HMAC Algorithm
- DAA Algorithm

Algorithm

1. Read input file (exam paper).
2. Generate hash value using SHA-256.
3. Store original hash.
4. Modify file (if tampered).
5. Generate new hash.
6. Compare hash values.
7. Detect integrity violation

Result

Thus, **hash functions** were successfully implemented to verify **data integrity** in an online examination system, ensuring tamper detection.

Conclusion

This experiment helps students to:

- Understand integrity protection
- Learn cryptographic hashing
- Prepare for digital signature implementation

Experiment No: 5

Title:

Implement the Digital Signature algorithm using RSA approach or DSA approach

Aim:

To implement digital signature creation and verification using RSA and to ensure authentication, integrity, and non-repudiation in an Online Examination System.

Capstone Scenario – Online Examination System

In an online examination system:

- Students must be assured that exam papers are genuine
- Authorities must prevent denial of exam paper creation
- Tampered exam papers must be detected

In this experiment:

- The exam authority digitally signs the question paper
- Students verify the digital signature before accessing the exam

This simulates secure exam paper authentication and non-repudiation.

Theory:

- RSA Approach
- DSS Approach
- Elgamal Digital Signature Scheme

Algorithm

Algorithm – Digital Signature using RSA

1. Generate RSA key pair.
2. Read exam paper file.
3. Generate hash of file.
4. Encrypt hash using private key.
5. Send file and signature.
6. Receiver decrypts signature using public key.
7. Compare hashes.
8. Verify authenticity.

Result

Thus, the digital signature algorithm using RSA was successfully implemented to provide authentication, integrity, and non-repudiation in an online examination system.

Conclusion

This experiment helps students to:

- Understand digital signatures
- Learn authentication mechanisms
- Prepare for SSL and certificate-based security